



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**Faculty of
Computing**

LAB ASSIGNMENT 2

INDUSTRY PROJECT MODULE PROPOSAL

GROUP NAME:

DEVSYNC

PROJECT TITLE:

eHadir

INDUSTRY PARTNER NAME:

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1.0 Project Background

Currently, lecturers and administrative staff face many challenges when managing student attendance and class schedules. They rely on a scattered system that uses multiple different applications, such as Microsoft Excel, Power Automate, and Power BI. This setup is very difficult to use and maintain because many staff members do not have the advanced technical expertise needed to fix complex software connections if something breaks. Additionally, the university faces problems with limited third-party software subscriptions. Since everything is separated across different apps, simple daily tasks take up too much time and effort.

The manual process creates several specific problems. First, it is highly inefficient for lecturers to track daily student attendance such as marking who is present, absent, on medical leave, or official leave. Second, when students are frequently absent or have misconduct issues, it is hard to record these discipline problems quickly. Third, managing schedules is confusing as administrative staff struggle to upload complex timetable data and lecturers do not have a quick way to view their teaching slots or easily book replacement classes when needed. Finally, the Head of Program does not have a simple way to view overall attendance statistics, percentages and trends.

To solve all of these problems, our group proposes the development of a centralized Learning Management System that focuses entirely on a complete Attendance System workflow. We will build a mobile application using Flutter for the user interface and Firestore for a secure, centralized database. The app will work smoothly on both iOS and Android devices, making it easy for lecturers and staff to use it from their phones anywhere on campus. This system is designed specifically to support the daily academic operations for a capacity of under 800 users.

By combining all these necessary tasks into one single platform, this application will heavily reduce manual administrative paperwork. It will allow staff to upload timetables, enable lecturers to view their slots, mark attendance instantly, report discipline issues, and book replacement classes without hassle. At the same time, it will automatically generate clear reports and statistics for the management. In conclusion, this project will save time, prevent errors, and provide a highly accurate, automated environment for tracking student attendance.

2.0 Target Users

The application is designed exclusively for internal faculty and administrative use. Students do not have direct login access to this system. The primary target users include:

1. Lecturers:

- a. To record daily student attendance, report discipline issues for students like absence or misconduct.
- b. Reschedule or book replacement class

2. Head of Program:

- a. To show attendance statistics and summaries such as percentage and trends

3. Administrative Staff:

- a. To upload timetable data for students and lecturers
- b. To allow lecturer to reschedule or book replacement class

3.0 Mandatory Modules

Module	Description	Lead Member
Login	Authenticated entry into the app.	Afif Shaqir Irfan
Registration	New user sign-up with basic validation.	Afif Shaqir Irfan
Manage User	View, edit, or remove user accounts	Syahmi Faris

4.0 Industry Modules

Module	Module Name	Description
Module 1	Taking Attendance	Enables lecturers to mark student attendance for each class session, including statuses such as <i>present</i> , <i>absent</i> , <i>MC (medical leave)</i> , and <i>CK (official leave)</i> . This module ensures accurate and real-time attendance tracking.
Module 2	Report Discipline Issue	Allows lecturers to record and manage student discipline cases, including frequent absence, lateness, or misconduct. This helps in maintaining student behavioral records for further action.
Module 3	Reporting Module	Provides analytical reports on student attendance, including statistics such as attendance percentage, trends, and summaries to support decision-making by the Head of Program.
Module 4	Upload Time Schedule	Enables administrative staff to upload and manage timetable data, including class, subject, lecturer, and time details, ensuring accurate scheduling within the system.
Module 5	Showing Timetable Slot	Displays lecturer's teaching schedules in structured format and allows quick navigation to attendance marking for each scheduled class.
Module 6	Booking Module	Allows lecturers to reschedule classes or book replacement sessions, providing flexibility in managing teaching schedules.

5.0 Model Distribution Table

Module	Type	Difficulty	Lead Member
Login	Mandatory	 Low	Afif Shaqir Irfan
Registration	Mandatory	 Low	Afif Shaqir Irfan
Manage User	Mandatory	 Medium	Syahmi Faris
Module 1 - Taking Attendance	Industry	 Medium	Afiq Danish
Module 2 - Report Discipline Issue	Industry	 Medium	Afiq Danish
Module 3 - Reporting Module	Industry	 High	Adam
Module 4 - Upload Time Schedule	Industry	 Medium	Syahmi Faris
Module 5 - Showing Timetable Slot	Industry	 Low	Afif Shaqir Irfan
Module 6 - Booking Module	Industry	 High	Choh Jing Yi

6.0 Justification

The distribution of modules among the five team members was determined based on two primary considerations, technical complexity and fairness of workload.

Afif Shaqir Irfan is assigned three modules, namely Login, Registration, and Module 5 (Showing Timetable Slot). Although he carries the highest module count, all three are rated Low difficulty, meaning the combined effort remains comparable to teammates handling fewer but more complex modules. Login and Registration are grouped together as they share the same authentication logic and UI patterns, while Showing Timetable Slot is a read-only display module with minimal business logic, making the overall workload manageable. Syahmi Faris is assigned Manage User and Module 4 (Upload Time Schedule), both rated Medium difficulty. These two modules are grouped together as they are both administratively driven, involving data management, user role handling, and structured data input. The pairing is logical as the user data managed in Manage User directly supports the role-based access enforced when uploading timetable data. Afiq Danish is assigned Module 1 (Taking Attendance) and Module 2 (Report Discipline Issue), both rated Medium difficulty. These modules are highly correlated as discipline issues such as frequent absence are identified directly from attendance records. Grouping them under one member eliminates cross-member data dependencies and ensures a consistent data flow between both modules.

Adam is assigned solely to Module 3 (Reporting Module) which is rated High difficulty. Despite handling only one module, the analytical complexity of aggregating attendance data, computing statistics and rendering dynamic charts for the Head of Program justifies the singular assignment. The development effort required for this module alone is considered equivalent to two Medium-difficulty modules. Choh Jing Yi is assigned solely to Module 6 (Booking Module) which also rated as High difficulty. The module involves real-time room availability checking, conflict detection against uploaded schedule data and managing rescheduling workflows, all of which demand complex state management and deep integration with the schedule data layer. A dedicated single-module assignment ensures sufficient focus and development time for this high-complexity deliverable.

7.0 Conclusion

In conclusion, the DevSync team is committed to delivering a centralized Attendance Management System that directly addresses the operational inefficiencies currently faced by the stakeholder in their day-to-day academic workflows that involves academic staff, lecturers and head of programs. By leveraging Flutter for cross-platform mobile development and Firestore as the backend data solution, the proposed system will unify processes that were previously scattered across multiple disconnected tools into an application. The modules outlined in this proposal include user authentication, attendance tracking, discipline reporting, timetable management, analytical reporting, and class booking collectively represent a comprehensive solution designed with real user needs at its core. With the tasks split equally in responsibilities among all five team members and with development progress to be actively monitored through NexLab on NexScholar, the team is confident in its ability to meet the expectations set by the industry partner and deliver a high-quality product by the final presentation in Week 14.